

Question 1 Use a SEPARATE Writing Booklet.**Marks**

- (a) Differentiate $x^2 \sin^{-1} x$. 2
- (b) Find the value of k if $x + 3$ is a factor of $P(x) = 2x^3 - 5kx + 9$. 2
- (c) The interval AB has end points $A(-3, 5)$ and $B(3, 2)$. Find the coordinates of the point P which divides the interval AB externally in the ratio $2 : 5$. 2
- (d) Find the acute angle, to the nearest degree, between the lines $x + y = 5$ and $2y = 3x + 5$. 2
- (e) Find $\lim_{\theta \rightarrow 0} \left(\frac{\sin^2 \theta}{\theta} \right)$. 2
- (f) Use the table of standard integrals to find the exact value of 2

$$\int_0^{\frac{3}{2}} \frac{dx}{\sqrt{x^2 + 4}}.$$

Question 2 Use a SEPARATE Writing Booklet.

Marks

- (a) Find the quotient, $Q(x)$, and the remainder, $R(x)$, when the polynomial $P(x) = 2x^4 - 3x^3 + 2x + 1$ is divided by $x^2 + 2x - 1$. **3**
- (b) Prove the identity $\frac{1 - \tan^2 A}{1 + \tan^2 A} = \cos 2A$. **2**
- (c) Find the value of x if $\frac{d}{dx} \left(\frac{x+2}{\sqrt{x-1}} \right) = 0$. **3**
- (d) Use the substitution $u = x - 1$ to evaluate $\int_2^5 \frac{x+1}{\sqrt{x-1}} dx$. **4**

Question 3 Use a SEPARATE Writing Booklet.

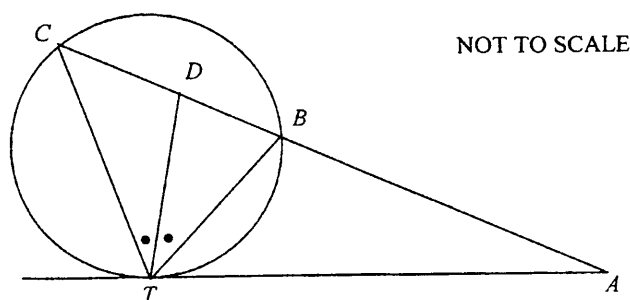
Marks

- (a) Solve $\frac{x}{x^2 - 4} \leq 0$. **3**
- (b) Solve $|x - 3| \leq 2x + 1$ **3**
- (c) Consider the function $f(x) = x \log_e x$.
- (i) Show that $y = f(x)$ has a minimum turning point at $(\frac{1}{e}, -\frac{1}{e})$. **4**
- (ii) Hence sketch the curve of $y = f(x)$ for $x \geq \frac{1}{e}$. **1**
- (iii) On the same set of axes as part (ii), draw the graph of the inverse function of $y = f(x)$, $x \geq \frac{1}{e}$. **1**

Question 4 Use a SEPARATE Writing Booklet.

Marks

(a)



3

TA is a tangent to a circle. Line $ABDC$ intersects the circle at B and C . Line TD bisects angle BTC .

Prove $AT = AD$.

- (b) $P(2ap, ap^2)$ is any point on the parabola $x^2 = 4ay$. The line l is parallel to the tangent at P and passes through the focus, S , of the parabola.

i) Find the equation of the line l .

1

ii) The line l intersects the x -axis at the point Q . Find the coordinates of the midpoint, M , of the interval QS .

2

iii) What is the equation of the locus of M ?

1

- (c) Equipment being delivered by a parachute drop is falling at a speed of $v \text{ m s}^{-1}$. When the parachute opens, the equipment is falling at 50 m s^{-1} , and thereafter its acceleration is given by $\frac{dv}{dt} = k(2 - v)$, where k is a constant.

(i) Show that this equation for $\frac{dv}{dt}$ is satisfied by $v = 2 + Ae^{-kt}$, where A is a constant.

1

(ii) Find the value of A .

1

(iii) One second after the parachute opens, the speed of the equipment has fallen to 35 m s^{-1} . Determine the value of k correct to 4 decimal places.

2

(iv) After a period of time, the equipment continues to fall with a speed which is very nearly constant, and which is called the "terminal speed". Find the terminal speed for this particular parachute drop.

1

Question 5 Use a SEPARATE Writing Booklet.

Marks

(a) (i) Express $\sqrt{3} \cos 2t - \sin 2t$ in the form $R \cos(2t + \alpha)$, where $0 < \alpha < \frac{\pi}{2}$. **2**

(ii) Hence or otherwise find all positive solutions of $\sqrt{3} \cos 2t - \sin 2t = 0$. **2**

(b) A particle moves in straight line and is x metres from a fixed point O after t seconds, where $x = 5 + \sqrt{3} \cos 2t - \sin 2t$.

Note The results of part (a) may be useful in answering this part.

(i) Prove that the acceleration of the particle is $-4(x - 5)$. **2**

(ii) Between which two points does the particle oscillate? **1**

(iii) At what time does the particle first pass through the point $x = 5$? **1**

(c) Use mathematical induction to prove that **4**

$$1^2 + 3^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$$

for all integers $n = 1, 2, 3, \dots$.

Question 6 Use a SEPARATE Writing Booklet.**Marks**

- (a) When a particle moving in a straight line has displacement x metres from a fixed point O , its acceleration in metres per second per second is given by $\ddot{x} = \sqrt{3x+4}$.
- (i) Show that $v^2 = \frac{4}{9}(3x+4)^{\frac{3}{2}} + c$, where v is the velocity of the particle in metres per second, and c is a constant. **1**
- (ii) Given that the particle starts from rest at O , evaluate c . **1**
- (iii) Explain why the motion of the particle is always in the positive direction. **1**
- (b) A spherical map of the earth is being inflated at a constant rate of $25\text{cm}^3\text{s}^{-1}$. Find the rate at which the length of the equator is changing when the radius is 10cm . **3**
- (c) (i) By using graphs or otherwise, show that the curves $y = \ln x$ and $y = 2 - x$ have a point of intersection for which the x coordinate is close to 1.5 . **1**
- (ii) Use one application of Newton's method to find a better approximation for the x coordinate of this point of intersection, correct to two decimal places. **2**
- (d) A solid is formed by rotating the curve $y = 1 + \sqrt{2} \cos x$, between $x = 0$ and $x = \frac{\pi}{4}$, about the x -axis. Find the exact volume of the solid. **3**

Question 7	Use a SEPARATE Writing Booklet.	Marks
(a)	Write down the last digit of the expansion of 7^{2002} . (You are only required to write the units digit).	2
(b)	(i) Differentiate $\sin^{-1} x + \cos^{-1} x$	1
	(ii) Hence, or otherwise explain why $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$	1
	(iii) Find the exact values of x and y which satisfy the simultaneous equations	3
	$\sin^{-1} x - \cos^{-1} y = \frac{\pi}{12}; \quad \cos^{-1} x + \sin^{-1} y = \frac{5\pi}{12}$	
(c)	A vertical tower subtends angles α and β respectively at two points A and B in a horizontal plane through the base of the tower. A is due south of the tower and B is due west.	
	(i) Draw a diagram to illustrate this information.	1
	(ii) Show that the cosine of the angle subtended at the top of the tower by the line AB is $\sin \alpha \sin \beta$.	4

END OF EXAMINATION

TRIAH EXAM EXT 1 2002.

Q1. a) $y = x^2 \sin^{-1} x$

$$\frac{dy}{dx} = \frac{x^2}{\sqrt{1-x^2}} + 2x \sin^{-1} x$$

1 for use of product rule

1 for correct answer.

b) $P(x) = 2x^3 - 5kx + 9$

$$P(-3) = 0$$

$$\therefore 0 = 2(-3)^3 - 5k(-3) + 9$$

1 for correct subst at this line

$$-54 + 15k + 9 = 0$$

$$15k = 45$$

$$k = 3$$

1 for correct answer

c) A(-3, 5) B(3, 2)

$$-2:5$$

$$x = \frac{-2 \times 3 + 5 \times -3}{-2 + 5}$$

$$= \frac{-6 - 15}{3}$$

$$= \frac{-21}{3}$$

$$= -7$$

$$\therefore P \rightarrow (-7, 7)$$

$$y = \frac{-2 \times 2 + 5 \times 5}{3}$$

$$= \frac{-4 + 25}{3}$$

$$= \frac{21}{3}$$

$$= 7$$

1 mark for each co-ordinate

d) $x + y = 5$

$$y = 5 - x$$

$$m_1 = -1$$

$$2y = 3x + 5$$

$$y = \frac{3}{2}x + \frac{5}{2}$$

$$m_2 = \frac{3}{2}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-1 - \frac{3}{2}}{1 - \frac{3}{2}} \right|$$

$$= 5$$

$$\text{acute } \angle \theta = 79^\circ \text{ to nearest degree}$$

1 for correct subst in formula

1 for answer

Allow 1st mark if 1 error in finding grad

e) $\lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\theta} = \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \cdot \frac{\sin \theta}{\theta} \cdot \theta$

$$= \lim_{\theta \rightarrow 0} \theta = 0$$

1 mark

1 mark

f) $\int_0^{3/2} \frac{dx}{\sqrt{x^2 + 4}} = \left[\ln(x + \sqrt{x^2 + 4}) \right]_0^{3/2}$

$$= \ln 2$$

1 mark

1 mark

$$\begin{array}{r}
 2x^2 - 7x + 16 \\
 2x^2 + 2x - 1 \overline{) 2x^4 - 3x^3 + 2x + 1} \\
 \underline{2x^4 + 4x^3 - 2x^2} \\
 -7x^3 + 2x^2 + 2x + 1 \\
 \underline{-7x^3 - 14x^2 + 7x} \\
 16x^2 - 5x + 1 \\
 \underline{16x^2 + 32x - 16} \\
 -37x + 17
 \end{array}$$

1 mark each step
in the division

$$Q(x) = 2x^2 - 7x + 16, R(x) = -37x + 17$$

b) To prove that $\frac{1 - \tan^2 A}{1 + \tan^2 A} = \cos 2A$.

$$\frac{1 - \tan^2 A}{1 + \tan^2 A} = \frac{1 - \tan^2 A}{\sec^2 A}$$

1 mark.

$$= \cos^2 A - \frac{\sin^2 A}{\cos^2 A} \times \cos^2 A$$

$$= \cos^2 A - \sin^2 A$$

$$= \cos 2A.$$

1 mark.

c) $\frac{d}{dx} \left(\frac{x+2}{\sqrt{x-1}} \right) = 0$

$$\frac{\sqrt{x-1} \times 1 - (x+2) \times \frac{1}{2} (x-1)^{-\frac{1}{2}}}{x-1} = 0$$

$$\frac{2(x-1) - (x+2)}{2(x-1)\sqrt{x-1}} = 0$$

$$\therefore 2x - 2 - x - 2 = 0$$

$$\underline{x = 4}$$

1 for correct derivative.

1 for simplification

1 for answer

d) $\int_2^5 \frac{x+1}{\sqrt{x-1}} dx$

$$= \int_1^4 \frac{u+2}{\sqrt{u}} du$$

$$= \int_1^4 \left(\sqrt{u} + 2u^{-\frac{1}{2}} \right) du$$

$$= \left[\frac{2}{3} u^{\frac{3}{2}} + 4u^{\frac{1}{2}} \right]_1^4$$

$$= \left(\frac{2}{3} \times 8 + 4 \times 2 \right) - \left(\frac{2}{3} + 4 \right)$$

$$= \frac{14}{3} + 4$$

$$= 8 \frac{2}{3}$$

$$u = x - 1$$

$$du = dx$$

$$x = 5, u = 4$$

$$x = 2, u = 1$$

1 for correct s

1 for integral

1 for subst of
correct limit

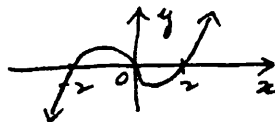
1 for answer

Q3. a) $\frac{x}{x^2-4} \leq 0$

$$x(x^2-4) \leq 0$$

$$x(x-2)(x+2) \leq 0$$

$$\therefore x < -2, 0 \leq x < 2$$

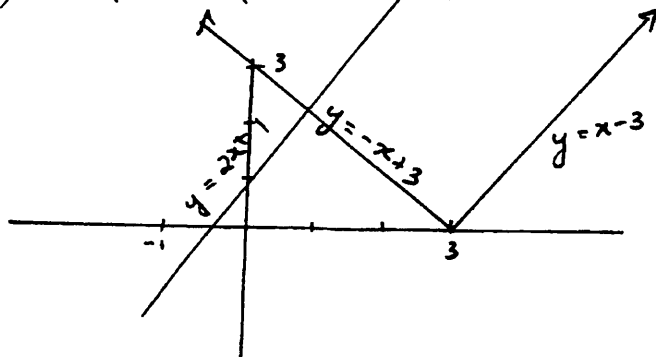


1 mark.

2 marks

Award 1 mark
for $x \leq -2, 0 \leq x < 2$

b) $|x-3| \leq 2x+1$



1 mark each graph
1 correct solution

$$2x+1 = -x+3$$

$$3x = 2$$

$$x = \frac{2}{3}$$

$$\therefore |x-3| \leq 2x+1$$

$$x \geq \frac{2}{3}$$

c) i) $y = x \log_e x$

$$x = \frac{1}{e}, y = \frac{1}{e}x - 1$$

$$\frac{dy}{dx} = 1 + \log_e x$$

$$\frac{d^2y}{dx^2} = \frac{1}{x}$$

For a turning pt. $\frac{dy}{dx} = 0$

$$\therefore 1 + \log_e x = 0, \log_e x = -1$$

$$\therefore x = e^{-1}$$

$$x = \frac{1}{e}, \frac{d^2y}{dx^2} = e > 0 \therefore \text{Min tp at } \left(\frac{1}{e}, -\frac{1}{e}\right)$$

1 mark for deriva

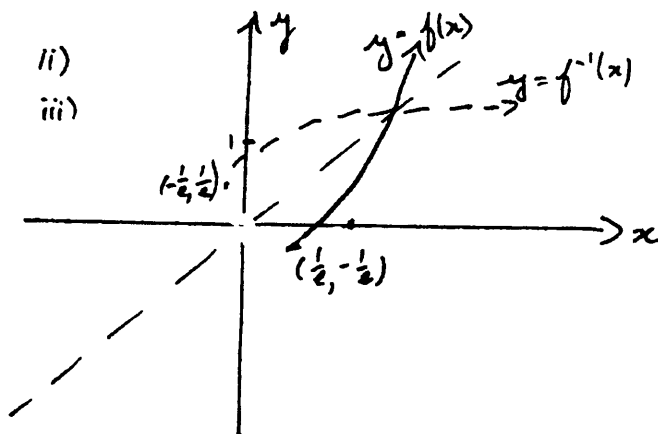
1 mark for $x = e$

1 mark for testing
for minima.

1 mark for y-co-ord

ii)

iii)



1 for each graph
in (ii) & (iii)

1 mark

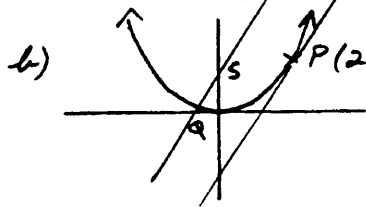
(given)

1 mark

$$= \angle ATB + \angle DTB$$

$$= \underline{LDTA}$$

1 mark.



1 mark.

1 mark

/ mark

1 mark.

c) i)

$$A e^{-k t} = v - 2$$

$$\begin{aligned} \frac{dI}{dt} &= -kAe \\ &= k(-Ae^{-kt}) \end{aligned}$$

$$= k(-Ae^{-kt})$$

$$= k(2 - v)$$

1 mark

A = 48

1 mark.

(mark -

$$+k = -\ln \frac{33}{49} = 0.3747 \text{ to 4dp. } 1 \text{ mark}$$

iv)

/mark

Terminal velocity is 2 m/sec

$$5a) i) \sqrt{3} \cos 2t - \sin 2t = R \cos(2t + \alpha) \\ = R \cos 2t \cos \alpha - R \sin 2t \sin \alpha$$

$$\therefore R \cos \alpha = \sqrt{3}, R \sin \alpha = 1$$



1 mark for R

1 mark for α

$$\therefore R = 2, \alpha = \frac{\pi}{6}$$

$$\therefore \sqrt{3} \cos 2t - \sin 2t = 2 \cos(2t + \frac{\pi}{6})$$

$$ii) \sqrt{3} \cos 2t - \sin 2t = 0$$

$$2 \cos(2t + \frac{\pi}{6}) = 0$$

$$2t + \frac{\pi}{6} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \dots$$

$$2t = \frac{\pi}{3}, \frac{4\pi}{3}, \frac{7\pi}{3}, \frac{10\pi}{3}, \dots$$

$$t = \frac{\pi}{6}, \frac{2\pi}{3}, \frac{7\pi}{6}, \frac{5\pi}{3}, \dots$$

$$\text{OR } 2t + \frac{\pi}{6} = (2n+1)\frac{\pi}{2}$$

1 mark for.

1 mark answer.

$$\text{OR } n\frac{\pi}{2} + \frac{\pi}{6} \text{ is equivalent.}$$

$$b) i) x = 5 + \sqrt{3} \cos 2t - \sin 2t$$

$$\dot{x} = -2\sqrt{3} \sin 2t - 2 \cos 2t$$

$$\ddot{x} = -4\sqrt{3} \cos 2t + 4 \sin 2t$$

$$= -4(\sqrt{3} \cos 2t + \sin 2t)$$

$$= -4(x-5)$$

1 for differentiations

1 for substitution

$$ii) \text{ Centre of oscillation } 5, \text{ amplitude } 2$$

$$\therefore \text{ Oscillates between } 3 \text{ \& } 7$$

1 mark

$$iii) \text{ First passes } x=5 \text{ after } \frac{\pi}{6} \text{ secs}$$

1 mark

$$c) n=1 \quad 1^2 + 3^2 + \dots (2n-1)^2 = 1^2 \quad \frac{1}{3}n(2n-1)(2n+1) = \frac{1}{3} \times 1 \times 3 = 1$$

$$\therefore 1^2 + 3^2 + \dots (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1) \text{ when } n=1 \quad 1 \text{ mark.}$$

$$\text{Assume } 1^2 + 3^2 + \dots (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1) \text{ when } n=k, k \geq 1, n \text{ an integer}$$

$$\text{ii assume } 1^2 + 3^2 + \dots (2k-1)^2 = \frac{1}{3}k(2k-1)(2k+1)$$

$$\text{When } n=k+1, \quad 1^2 + 3^2 + \dots (2n-1)^2 = 1^2 + 3^2 + \dots (2k-1)^2 + (2k+1)^2 \\ = \frac{1}{3}k(2k-1)(2k+1) + (2k+1)^2 \text{ from assumption} \\ = \frac{1}{3}(2k+1)[2k^2 - k + 6k + 3] \\ = \frac{1}{3}(2k+1)(2k+3)(k+1) \quad 2 \text{ mark.} \\ = \frac{1}{3}(2n-1)(2n+1)n \text{ when } n=k+1$$

$$\text{If } 1^2 + 3^2 + \dots (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1) \text{ when } n=k$$

$$\text{then } 1^2 + 3^2 + \dots (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1) \text{ when } n=k+1$$

$$\text{Since } 1^2 + 3^2 + \dots (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1) \text{ when } n=1$$

$$\text{then } 1^2 + 3^2 + \dots (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1) \text{ when } n=2, 3, 4, \dots$$

$$\text{Therefore } 1^2 + 3^2 + \dots (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1) \text{ for all +ve integral values of } n$$

1 mark

Q6. a) $\ddot{x} = \sqrt{3x+4}$
 $\frac{d}{dx}(\frac{1}{2}v^2) = \sqrt{3x+4}$
 $\frac{1}{2}v^2 = \frac{1}{3} \int 3(3x+4)^{\frac{1}{2}} dx$
 $= \frac{1}{3} (3x+4)^{\frac{3}{2}} \times \frac{2}{3} + C$
 $v^2 = \frac{4}{9} (3x+4)^{\frac{3}{2}} + C$
 ii) $x=0, v=0, \therefore 0 = \frac{4}{9} \times 4^{\frac{3}{2}} + C$
 $C = -\frac{64}{27}$

1 mark.

1 mark.

iii) Since $3x+4 \geq 0, x \geq -\frac{4}{3}$. Motion begins at 0 $\therefore$ must go in +ve direction
 OR accel is +ve $\therefore$ vel increases etc

1 mark.

b) $\frac{dV}{dt} = 25$ $\frac{dE}{dt} = ?$
 $V = \frac{4}{3} \pi r^3$ $E = 2\pi r$
 $\frac{dV}{dr} = 4\pi r^2$ $\frac{dE}{dr} = 2\pi$

1 for correct info.

$$\frac{dE}{dt} = \frac{dE}{dr} \times \frac{dr}{dV} \times \frac{dV}{dt}$$

$$= 2\pi \times \frac{1}{4\pi r^2} \times 25$$

1 for this line

$r=10, \frac{dE}{dt} = \frac{1}{200} \times 25$
 $= \frac{1}{8}$

$\therefore$ Equator is increasing at $\frac{1}{8}$ cm/sec.

1 for answer.

c) i) $y = \ln 1.5 = 0.4$ $y = 2 - 1.5 = 0.5$

$\therefore$ Pt of intersection is close to $x=1.5$

1 mark.

ii) $\ln x - 2 + x = 0$

Let $x_1 = 1.5$

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$f'(x) = \frac{1}{x} + 1$

$f'(1.5) = \frac{2}{3} + 1 = \frac{5}{3}$

$= 1.5 - \frac{-0.1}{\frac{5}{3}}$

$= 1.5 + 0.1 \times \frac{3}{5}$

$= 1.5 + 0.06$

$= 1.56$

d) $V = \pi \int_0^{\pi/4} (1 + \sqrt{2} \cos x)^2 dx = \pi \int_0^{\pi/4} (1 + 2\sqrt{2} \cos x + 2 \cos^2 x) dx$

$= \pi [x + 2\sqrt{2} \sin x]_0^{\pi/4} + \pi \int_0^{\pi/4} (\cos 2x + 1) dx$

$= \pi [x + 2\sqrt{2} \sin x + \frac{1}{2} \sin 2x + x]_0^{\pi/4}$

$= \pi [\frac{\pi}{2} + 2 + \frac{1}{2}]$

$\text{Vol} = \frac{\pi}{2} (\pi + 5) \text{ m}^3$

1 for setup for $2 \cos^2 x$

1 for integration

1 for answer

Q7. a) Last digits are 7, 9, 3, 1, 7, 9, 3...

1 mark

$2002 \div 4$ leaves a remainder of 2.

$\therefore$ Last digit is 9

1 mark.

b) i) $\frac{d}{dx}(\sin^{-1}x + \cos^{-1}x) = \frac{1}{\sqrt{1-x^2}} + \frac{-1}{\sqrt{1-x^2}} = 0$

1 mark.

ii) Since the derivative is zero, $\sin^{-1}x + \cos^{-1}x = C$

Let $x = 0$, $\sin^{-1}0 + \cos^{-1}0 = 0 + \frac{\pi}{2} \therefore C = \frac{\pi}{2}$

1 mark.

i.e. $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$

iii) $\sin^{-1}x - \cos^{-1}y = \frac{\pi}{12}$

Let $\alpha = \sin^{-1}x \therefore \sin \alpha = x$

$\beta = \cos^{-1}y \quad \cos \beta = y$

$$\begin{aligned} \cos^{-1}x + \sin^{-1}y &= \frac{5\pi}{12} \\ \frac{\pi}{2} - \sin^{-1}x + \frac{\pi}{2} - \cos^{-1}y &= \frac{5\pi}{12} \\ \sin^{-1}x + \cos^{-1}y &= \frac{7\pi}{12} \end{aligned}$$

1 mark.

$\therefore \alpha - \beta = \frac{\pi}{12}$

$\alpha + \beta = \frac{7\pi}{12}$

$2\alpha =$

$\frac{8\pi}{12} = \frac{2\pi}{3}$

$\alpha = \frac{\pi}{3}$

$\therefore \beta = \frac{\pi}{4}$

$\therefore x = \sin \frac{\pi}{3}$

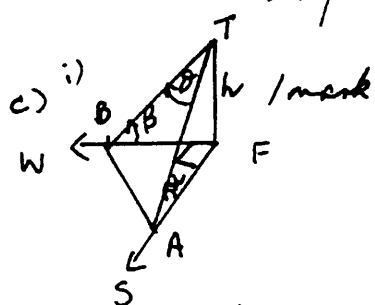
1 mark α ,

$= \frac{\sqrt{3}}{2}$

$y = \cos \frac{\pi}{4}$

$= \frac{1}{\sqrt{2}}$

1 mark x, y



ii) $\sin \beta = \frac{h}{BT}$

$BT = \frac{h}{\sin \beta}$

$\sin \alpha = \frac{h}{AT}$

$AT = \frac{h}{\sin \alpha}$

$\tan \beta = \frac{h}{BF}$

$BF = h \cot \beta$

$\tan \alpha = \frac{h}{AF}$

$AF = h \cot \alpha$

1 mark

In $\triangle BFA \quad AB^2 = BF^2 + AF^2$

$= h^2 \cot^2 \beta + h^2 \cot^2 \alpha$

1 mark

In $\triangle BTA$ by cosine rule $\cos \theta = \frac{BT^2 + AT^2 - AB^2}{2BT \times AT}$

$= \frac{\frac{h^2}{\sin^2 \beta} + \frac{h^2}{\sin^2 \alpha} - \frac{h^2}{\tan^2 \beta} - \frac{h^2}{\tan^2 \alpha}}{2 \times \frac{h}{\sin \beta} \times \frac{h}{\sin \alpha}}$

1 mark using Cosine Rule

$= \frac{\sin^2 \alpha + \sin^2 \beta - \cos^2 \beta \sin \alpha - \cos^2 \alpha \sin \beta}{2 \sin \alpha \sin \beta}$

1 mark for correct answer

$= \frac{\sin^2 \alpha (1 - \cos^2 \beta) + \sin^2 \beta (1 - \cos^2 \alpha)}{2 \sin \alpha \sin \beta}$

$= \frac{\sin^2 \alpha \sin^2 \beta + \sin^2 \beta \sin^2 \alpha}{2 \sin \alpha \sin \beta}$

$= \sin \alpha \sin \beta$